

Sentence-Final Particles in Late Archaic Chinese

1. Introduction

The split-CP hypothesis proposed by Rizzi (1997), Cinque (1999), Cinque and Rizzi (2008) etc. has successfully plotted a map of the left periphery by using data from different modern languages. This paper extends this cartographic approach to historical linguistics, specifically to the sentence final particles (SFP) in late Archaic Chinese (LAC) (5th– 3rd c. BC).

Previously, Djamouri and Paul (2019) and Pan (2021) proposed a rough hierarchy of SFPs in LAC that involves yě (也), hū (乎), and zāi (哉). Similar to modern Chinese, it comprises three domains: low C (tense-related), medium C (force-related), and high C (attitude-related). (1) is an example in Pan (2021), where the FiniteP belongs to low C, iForceP to medium C, and –SQP and AttP to high C respectively.

(1) 我王者也乎哉？！

[AttitudeP [–SQP [iForceP [FiniteP [TP Wǒ wáng-zhě] yě] hū]] zāi]
1SG king YE HU ZAI

‘Oh, how come I am the king?’

→ ‘I am absolutely not the king!’

(國語 Guoyu, Spring and Autumn period)

(Pan 2021)

Nonetheless, as Pan (2021) points out, a detailed analysis of more SFPs in Archaic Chinese is still called for. This study aims to bridge this gap by identifying the discourse functions and functional positions of eight common SFPs in LAC: yě (也), yǐ₁ (已), yǐ₂¹ (矣), ěr (耳), hū (乎), yǔ (與), fū (夫), and zāi (哉).

In this paper, it is assumed that each SFP heads their own projection. The issue of headedness (whether the projections are head-initial or head-final) will not be discussed here.

2. Methods

This paper will first investigate the meanings or discourse functions of the SFPs to assign different functional projections to them. A predicted hierarchy can then be formed according to these projections. The resulting predictions will be tested by examining instances of collocation of SFPs in LAC. Data from the Analects in the Academic Sinica Ancient Chinese Corpus will be used to represent LAC. Translations from Lau (2014 [2000]) will be used. Searches of all mathematically possible SFP combinations are done in the corpus. Given that the characters used to represent the above SFPs may have other uses (e.g., yǐ (已) could be an adverb meaning ‘already’ in LAC other than functioning as an SFP), the search results will be examined carefully and only sentences with the target SFPs will be counted as examples. If any examples contradicting our proposed hierarchy are found, the hierarchy will be adjusted.

3. Core projections

¹ To differentiate between the two homophones 已 and 矣, 已 is labelled as yǐ₁ and 矣 as yǐ₂.

3.1. Yě (也) as FinP head

Previous studies such as Djamouri and Paul (2019), Wang (2023) point out that yě asserts the denoted proposition, i.e., the TP to which it attaches. (2) is an example.

(2) 見義不為，無勇也。

jiàn yì bù wéi wú yǒng yě

See right not do no courage YE

‘Seeing what ought to be done, to leave it undone shows a lack of courage.’ (Lau 2014 [2000]: 17)

In (2), yě asserts the proposition “it is a lack of courage”. However, yě having a relevancy function might be a more parsimonious explanation compared to having an asserting function. The author proposes that yě in LAC is semantically parallel to *ge3* in Cantonese. According to Sybesma and Li’s (2007) analysis on Cantonese *ge3*, instead of functioning as an assertion marker, *ge3* expresses that “the statement to which it is added is highly relevant to the current conversation”. The effect of assertion, focus, or emphasis is actually a side-effect of the high relevancy function of *ge3*. The present study suggests that yě in LAC functions in a similar way. Yě in (2) relates “lacking courage” to the preceding topic such that “lacking courage” is the utterer’s comment on “not taking action when faced with what is right”. This explains why yě is always used in equational sentences, as in (3).

(3) 政者，正也。

zhèng zhě zhèng yě

Govern TOP correct YE

‘Government (*cheng*) is being correct (*cheng*).’ (Lau 2014 [2000]: 115)

In (3), yě relates “to be correct” to the preceding topic “to govern”. Under this proposal, the equational meaning is only a side-effect of the relevancy function of yě.

As indicated, yě links a sentence to the preceding conversational context. On the other hand, a finite sentence is a sentence which is able to temporally anchor onto the context (Sybesma, 2004). Accordingly, yě can be analysed as the head of the FinP, relating the sentence to the preceding context by indicating that it shares the same temporal reference. This analysis matches the analysis adopted by Pan (2021) as in (1).

3.2. Yǐ (已) as S.AspP head

Yǐ₁ denotes a change of state, as in (4).

(4) 可謂好學也已。

kě wèi hǎo xué yě yǐ

Can describe like learn YE YI

‘Such a man can be described as eager to learn.’ (Lau 2014 [2000]: 7)

In (4), $yǐ_1$ signals the state of ‘can describe’. It takes scope over the modal verb ‘can’ yielding $yǐ_1 > \text{can}$, expressing the meaning ‘such a man can now be described as eager to learn’ (changing from the state ‘cannot be described as such’). Similar to modern Mandarin *le*, $yǐ_1$ is analysed as the head of S.AspP taking scope over the sentence.

It is observed that $yǐ_1$ always clusters with other SFPs in the Analects. Sentences with $yǐ_1$ as the only SFP are not found in the Analects. It is not clear whether this is a mere coincidence or there is a grammatical restriction underlying this observation. It is also worth noting that sentences with $yǐ_1$ as the only SFP are found in other LAC texts such as the Mencius.

3.3. $Yǐ_{2a}$ (矣) as S.AspP head²

$Yǐ_{2a}$ denotes a change of state, as in (5).

(5) 溫故而知新，可以為師矣。

wēn gù ér zhī xīn kěyǐ wéi shī yǐ
Revise old and know new can be teacher YI

‘A man is worthy of being a teacher who gets to know what is new by keeping fresh
in his mind what he is already familiar with.’ (Lau 2014 [2000]: 13)

In (5), $yǐ_{2a}$ signals the state of ‘can be’. It takes scope over the modal verb ‘can’ yielding $yǐ_1 > \text{can}$, expressing the meaning ‘the man can now be a teacher given the mentioned qualities’ (changing from the state ‘could not be a teacher yet’). Similar to $yǐ_1$, $yǐ_{2a}$ is analysed as the head of S.AspP. In addition, the fact that $yǐ_1$ and $yǐ_{2a}$ share high phonetic, syntactic, and semantic resemblance might suggest that they are cognates.³ This provides further supporting evidence for assigning $yǐ_{2a}$ as a S.AspP head.

3.4. Ěr (耳) as OnlyP head

Ěr can mark different parts of a sentence as an exclusive focus, as in (6).

(6) 前言戲之耳。

qián yán xì zhī ěr
Previous remark jest 3OBJ ER

‘My remark a moment ago was only made in jest.’ (Lau 2014 [2000]: 171)

In (6), ěr takes scope over the predicate ‘jest’ to express the meaning ‘I was only jesting’. Similar to *éryǐ* in Mandarin and *ze1/zaa3* in Cantonese, it can be analysed as an OnlyP (or FocP) in the CP layer to scope over a TP.

² As we will see later, $yǐ_2$ as S.AspP head will be labelled as $yǐ_{2a}$ and $yǐ_2$ as AttP head will be labelled as $yǐ_{2b}$.

³ Logographically, $yǐ_1$ is the upper part of $yǐ_2$, suggesting an etymological relationship between the two.

3.5. Hū₁ (乎) as iForceP head

Hū₁ can transform a declarative sentence into a polar question, as in (7).

(7) 與朋友交而不信乎？

yǔ péng yǒu jiāo ér bù xìn hū

With classmate friend associate and NEG trustworthy HU

‘In my dealings with my friends have I failed to be trustworthy in what I say?’ (Lau 2014 [2000]: 3)

If we take out the hū₁ in (7), it will become a declarative sentence ‘in my dealings with my friends I have failed to be trustworthy in what I say’. Therefore, hū₁ can be seen as a typing particle (as in the Clausal Typing Hypothesis in Cheng (1991)) that bears an inherent interrogative force. It should be analysed as the head of iForceP.

3.6. Yǔ₁ (與) as iForceP head

Yǔ₁ can also transform a declarative sentence into a polar question, as in (8).

(8) 女弗能救與？

nǚ fú néng jiù yǔ

2SG NEG can save YU

‘Can you not save the situation?’

(Lau 2014 [2000]: 19)

Similarly, if we take out the yǔ₁ in (8), it will become a declarative sentence ‘you cannot save the situation’. It is also a typing particle that bears an inherent interrogative force. Therefore, it should be analysed as the head of iForceP. The difference between hū₁ and yǔ is unknown. They can both appear in wh-questions and polar questions (both neutral polar and biased questions). Future studies can investigate the syntactic and semantic differences between these two interrogative particles.

3.7. Fū (夫) as AttP head

Fū is used as an exclamative particle that express the speaker’s mood and attitude, as in (9).

(9) 而今而後，吾知免夫！

ér jīn ér hòu wú zhī miǎn fū

Now and after 1SG know spare fu

‘Only now am I sure of being spared!’

(Lau 2014 [2000]: 69)

Therefore, fū can be analysed as AttP head. However, a more refined understanding in its semantic and discourse functions is unknown.

3.8. Zāi (哉) as AttP head

Zāi is also used as an exclamative particle that express the speaker's mood and attitude, as in (10).

- (10) 管仲之器小哉！
guǎn zhòng zhī qì xiǎo zāi
Kuanchung GEN vessel small ZAI
'Kuan Chung was, indeed, a vessel of small capacity.' (Lau 2014 [2000]: 25)

Therefore, zāi can be analysed as AttP head. However, a more refined understanding in its semantic and discourse functions is unknown.

3.9. Hū₂ (乎) as AttP head

Other than functioning as an iForceP head that types interrogative sentences, hū also expresses the speaker's mood and attitude in exclamative sentences, as in (11).

- (11) 惜乎！
xī hū
Pity HU
'What a pity!' (Lau 2014 [2000]: 83)

Therefore, hū can also be analysed as AttP head. For convenience, hū as iForceP head will be labelled as hū₁ and hū as AttP head will be labelled as hū₂. Although hū can also attach to longer sentences, it is observed that there is a tendency for hū₂ to be attached to short noun phrases and adjective phrases such as (13).

3.10. Yǐ_{2b} (矣) as AttP head

Other than functioning as a S.AsP head that denotes a change of state, the author judges that yǐ_{2b} sometimes expresses the speaker's mood and attitude in exclamative sentences, as in (12).

- (12) 吾不欲觀之矣。
wú bù yù guān zhī yǐ
1SG NEG want witness 3OBJ YI
'I do not wish to witness that part of the ti sacrifice [...]' (Lau 2014 [2000]: 21)

Liu (2007) also points out that yǐ₂ marks subjective judgements/evaluations based on the speaker's perception. Therefore, yǐ₂ can also be analysed as AttP head. For convenience, yǐ₂ as S.AsP head will be labelled as yǐ_{2a} and yǐ₂ as AttP head will be labelled as yǐ_{2b}.

3.11. Yǔ₂ (與) as AttP head

Other than functioning as an iForceP head that types interrogative sentences, yǔ also expresses the speaker's mood and attitude in exclamative sentences, as in (13).

- (13) 孝弟也者，其為仁之本與！
 xiào dì yě zhě qí wéi rén zhī běn yǔ !
 Filial piety YE TOP this COP benevolence GEN root YU
 ‘Being good as a son and obedient as a young man is, perhaps, the root of a man's character.’ (Lau 2014 [2000]: 3)

Therefore, yǔ can also be analysed as AttP head. For convenience, yǔ as iForceP head will be labelled as yǔ₁ and yǔ as AttP head will be labelled as yǔ₂.

4. Hierarchy

According to the above assignment of functional projections to these SFPs based on their functions, we can predict that they follow the order in (14) when they appear in clusters:

- (14) TP < FinP (yě) < S.AspP (yǐ₁, yǐ_{2a}) < OnlyP (ě) < iForceP (hū₁, yǔ₁) < AttP (yǐ_{2b}, hū₂, yǔ₂, fū, zāi)

In addition to this order, we assume that the pairs yǐ_{2a} and yǐ_{2b}, hū₁ and hū₂, yǔ₁ and yǔ₂ will not co-occur due to haplology. We will now check the instances where these SFPs co-occur in the form of clusters to see if they follow our predicted order.

4.1. Clusters conforming to our prediction

4.1.1. Yě (也) < yǐ (已), hū (乎), yǔ (與), yǔ₂ (與), zāi (哉)

Since yě is a FinP head, it is located below the rest of the SFPs and is predicted to precede every other SFP. As predicted, it is found that yě precedes yǐ₁, yǔ₁, yǔ₂, and zāi.

- (15) 可謂好學也已。
 kě wèi hǎo xué yě yǐ
 Can describe like learn YE YI
 ‘Such a man can be described as eager to learn.’ (Lau 2014 [2000]: 7)
- (16) 仲由可使從政也與？
 zhòngyóu kě shǐ cóng zhèng yě yǔ
 Chungyu can cause follow government YE YU
 ‘Is Chung Yu good enough to be given office?’ (Lau 2014 [2000]: 49)
- (17) 其回也與！
 qí huí yě yǔ
 This Hui YE YU
 ‘It is Hui, I suppose.’ (Lau 2014 [2000]: 83)
- (18) 吾豈匏瓜也哉？

wú qǐ páoguā yě zāi
1SQ how gourd YE ZAI

‘How can I allow myself to be treated like a gourd [...]?’ (Lau 2014 [2000]:

173)

Although possible in our model, no instance of yě preceding yǐ_{2a}, hū₁, ěr, yǐ_{2b}, hū₂, and fū is found. Note that theoretically possible combinations of SFPs in modern languages like Cantonese are not necessarily attested either. The underlying reasons are yet to be investigated.

4.1.2. Yǐ₁ (已) < yǐ_{2b} (矣), hū₁ (乎)

Since yǐ₁ is a S.AspP head, it is located above FinP and below OnlyP, iForceP, and AttP. Therefore, it is predicted that yǐ₁ does not precede FinP head yě and does not co-occur with S.AspP head yǐ_{2a}. As predicted, instances of yǐ₁ preceding yě are not attested. It is also found that yǐ₁ precedes yǐ_{2b} and hū₁, conforming to our prediction.

- (19) 始可與言《詩》已矣！
shǐ kě yǔ yán shī yǐ yǐ
Begin can with discuss The Odes YI YI
‘Only with a man like you can one discuss the Odes.’ (Lau 2014 [2000]: 7)
- (20) 斯謂之仁已乎？
sī wèi zhī rén yǐ hū
This describe 3OBJ benevolence YI HU
‘Can a man (already) be said to be benevolent (simply because he is loath to speak)?’ (Lau 2014 [2000]: 109)

In (19), the yǐ₂ is analysed as yǐ_{2b} but not yǐ_{2a} because two S.AspP heads will compete for the same position and should never co-occur. Moreover, yǐ_{2b} contributes the speaker’s attitude (exclamation) to the sentence, which cannot be carried by yǐ₁. Therefore, (19) is not an instance of co-occurrence of yǐ₁ and yǐ_{2a}.

Although possible in our model, no instance of yǐ₁ preceding ěr, yǐ₁, hū₂, yǐ₂, fū, and zāi is found.

4.1.3. Yǐ_{2a} (矣) < hū₁ (乎), hū₂ (乎)

Since yǐ_{2a} is a S.AspP head, it is located above FinP and below OnlyP, iForceP, and AttP. Therefore, it is predicted that yǐ₁ does not precede FinP head yě and does not co-occur with S.AspP head yǐ₁. As predicted, instances of yǐ₁ preceding yě are not attested. It is found that yǐ_{2a} precedes hū₁, conforming to our prediction.

- (21) 仁矣乎？
Rén yǐ hū

Yī₂ in (21) is a yǐ_{2a} but not a yǐ_{2b}. In (21), yǐ_{2a} expresses a change of state and the result of the premises. The previous premise mentioned before this sentence is that ‘Ling Yin Tsu-wen gave no appearance of pleasure when he was made the prime minister three times. Neither did he give any appearance of displeasure when he was removed from office three times. He always told his successor what he had done during his term of office.’ (Lau 2014 [2000]) (21) asks about the result of these conditions: whether he can now be said to be benevolent. Therefore, yǐ₂ is a S.AspP head but not an AttP head in (21). It follows that yǐ_{2a} can precede hū₁, since S.AspP < iForceP.

It is also found that yǐ_{2a} precedes hū₂, conforming to our prediction.

(22) 已矣乎！

yǐ yǐ hū

End YI HU

‘I suppose I should give up hope!’

(Lau 2014 [2000]: 153)

Yī₂ in (22) is a yǐ_{2a} but not a yǐ_{2b}. In (22), yǐ_{2a} expresses a change of state and the consequence of an observation. The observation (by Confucius) is ‘I have yet to meet the man who, on seeing his own errors, is able to take himself to task inwardly’ (Lau 2014 [2000]). Under this observation, (22) expresses his change of state: from having hope in humanity to giving up hope. Therefore, yǐ₂ is a S.AspP head but not an AttP head in (22). It follows that yǐ_{2a} can precede hū₂, since S.AspP < AttP.

Although possible in our model, no instance of yǐ_{2a} preceding ěr, yǔ₁, yǔ₂, fū, and zāi is found.

4.1.4. Absence of co-occurrence of ěr (耳) and other SFPs

Since ěr is an OnlyP head, it is located above S.AspP and below iForceP, and AttP. Therefore, it is predicted that yǐ₁ does not precede FinP head yě, and S.AspP heads yǐ₁ and yǐ_{2a}. However, ěr is not found to co-occur with any SFPs. In fact, only one instance of ěr is found in the Analects (i.e., (6)). This could be due to limited linguistic data in the Analects. Although our predictions are not attested, the absence of co-occurrence of ěr and other SFPs does not conflict with our model.

4.1.5. Hū₁ (乎) < zāi (哉)

Since hū₁ is an iForceP head, it is located above OnlyP and below AttP. Therefore, it is predicted that yǐ₁ does not precede FinP head yě, OnlyP head ěr, and S.AspP heads yǐ₁ and yǐ_{2a}. It is also predicted that hū₁ will not co-occur with yǔ₁ as they compete for the same position. These predictions are all borne out. It is further predicted that hū₁ can precede AttP heads yǐ_{2b}, yǔ₂, fū, zāi. It is found that hū₁ precedes zāi, conforming to our prediction.

- (23) 君子多乎哉？
 jūn zǐ duō hū zāi
 Gentleman many HU ZAI
 ‘Should a gentleman be skilled in many things?’ (Lau 2014 [2000]: 79)

Although possible in our model, no instance of hū₁ preceding yǐ_{2b}, yǔ₂, and fū is found.

4.1.6. Yǔ₁ (與) < zāi (哉)

Since yǔ₁ is an iForceP head, it is located above OnlyP and below AttP. Therefore, it is predicted that yǔ₁ does not precede FinP head yě, OnlyP head ěr, and S.AspP heads yǐ₁ and yǐ_{2a}. As predicted, instances of yǔ₁ preceding these SFPs are not attested. It is further predicted that yǔ₁ can precede AttP heads yǐ_{2b}, hū₂, fū, zāi. It is found that yǔ₁ precedes zāi, conforming to our prediction.

- (24) 可與事君也與哉？
 kě yǔ shì jūn yě yǔ zāi
 Can with serve lord YE YU ZAI
 ‘Is it really possible to work side by side with a mean fellow in the service of a lord?’ (Lau 2014 [2000]: 177)

Although possible in our model, no instance of yǔ preceding yǐ_{2b}, hū₂, and fū is found.

4.2. Problematic clusters

As we can see above, most observations conform to our prediction. Nonetheless, problematic clusters are found, as in (25) and (26) below.

- (25) 命矣夫！
 mìng yǐ fū
 Destiny YI FU
 ‘It must be destiny!’ (Lau 2014 [2000]: 49)
- (26) 難矣哉！
 nán yǐ zāi
 Difficult YI ZAI
 ‘[...] is sure to meet with difficulties’ (Lau 2014 [2000]: 179)

Yǐ₂ in (25) is a yǐ_{2b} but not a yǐ_{2a}. In (25), yǐ₂ only expresses the speaker (Confucius)’s despair. The context is that Confucius was visiting his student Po-niu who was ill and he believed that they were going to lose him. Therefore, he exclaimed ‘it must be destiny!’ (Lau 2014 [2000]). Yǐ₂ in (25) does not express any change of state. Therefore, yǐ₂ is an AttP head but not a S.AspP head in (25).

Similarly, $y\check{2}$ in (26) is a $y\check{2}_b$ but not a $y\check{2}_a$. In (26), $y\check{2}$ only expresses the speaker (Confucius)’s attitude and subjective judgement. In the context of (26), Confucius was commenting on ‘a man whose belly is full all day and does not put his mind to some use’ (Lau 2014 [2000]). Therefore, he commented that it is sure for such a person to meet with difficulties. $Y\check{2}$ in (26) expresses Confucius’ negative attitude on this type of people and does not express a change of state. Therefore, $y\check{2}$ is an AttP head but not a S.AspP head in (26).

However, if we analyse the $y\check{2}$ as AttP head $y\check{2}_b$ in (25) and (26), its co-occurrences with $f\bar{u}$ and $z\bar{a}i$ in these examples would violate the fact that SFPs should not co-occur if they compete for the same position, since $f\bar{u}$ and $z\bar{a}i$ are both AttP heads.

To solve this problem, I propose that there are two layers of AttP in LAC: AttP1 and AttP2, similar to Pan’s (2021) analysis of AttP in Mandarin. $Y\check{2}_b$ will occupy the AttP1 head while $h\bar{u}_2$, $f\bar{u}$, $y\check{2}_2$, and $z\bar{a}i$ will occupy the AttP2 head (since no co-occurrence of $h\bar{u}_2$, $f\bar{u}$, $y\check{2}_2$, and $z\bar{a}i$ is found). This allows $y\check{2}$ in (25) and (26) to be an AttP head.

For now, the following hierarchy of the eleven SFPs can be constructed.

- (27) [AttP2 [AttP1 [iForceP [OnlyP [S.AspP [FinP [TP ...] $y\check{e}$] $y\check{1}_1$, $y\check{2}_a$] $\check{e}r$] $h\bar{u}_1$, $y\check{u}_1$] $y\check{2}_b$] $h\bar{u}_2$, $y\check{2}_2$, $f\bar{u}$, $z\bar{a}i$]

This paper will not discuss in detail how the surface final order of the SFPs in (27) is derived since it is not the focus of this study. A possible account can be a Comp-to-Spec raising analysis: Assuming that the SFPs are head-initial, the SFPs are originally located on the left side of the TP (hence the left periphery). The TP then undergoes movement from the complement position to the specifier position of, say, FinP. If there is another SFP above FinP, e.g., the iForceP head $h\bar{u}_1$, the FinP will raise to the specifier position of iForceP, deriving the surface final order.⁴

Below is a table summarising the impossible and possible sequences predicted by this hierarchy, as well as the actual sequences observed.

Table 1. Summary of predicted sequences and observed sequences

		Fin	S.Asp		Only	iForce		Att1	Att2			
	A B	$y\check{e}$ (也)	$y\check{1}_1$ (已)	$y\check{2}_a$ (矣)	$\check{e}r$ (耳)	$h\bar{u}_1$ (乎)	$y\check{u}_1$ (與)	$y\check{2}_b$ (矣)	$h\bar{u}_2$ (乎)	$y\check{2}_2$ (與)	$f\bar{u}$ (夫)	$z\bar{a}i$ (哉)
Fin	$y\check{e}$ (也)											
S.Asp	$y\check{1}_1$ (已)	✓										
	$y\check{2}_a$ (矣)	X										

⁴ Regarding the motivation of movement, the comp-to-spec raising is used as a last resort to satisfy the EPP feature. For details, please refer to p. 207–211 in Pan (2021).

Only	ěr (耳)	X	X	X								
iForce	hū ₁ (乎)	✓	✓	✓	X							
	yǔ ₁ (與)	✓	X	X	X							
Att1	yǐ _{2b} (矣)	X	✓		X	X	X					
Att2	hū ₂ (乎)	X	X	✓	X		X	✓				
	yǔ ₂ (與)	✓	X	X	X	X		X				
	fū (夫)	X	X	X	X	X	X	✓				
	zāi (哉)	✓	X	X	X	✓	✓	✓				

A: preceding SFP; B: following SFP; greyed cells: theoretically impossible sequences; X: possible sequences but unobserved; ✓: possible and observed sequences

Note that the table applies not only to two-particle clusters, but also to n-particle clusters. It shows whether the SFPs in row A can precede the SFPs in column B in our model, and whether these sequences are observed. As predicted, no sequences in the greyed cells are observed.

5. Embeddability

Pan (2021) generalised a Subjectivity Scale Constraint, as shown below:

(28) Subjectivity Scale Constraint

The higher a functional projection is, the more subjective the interpretation of such a projection becomes, and the more difficult it is for such a projection to be embedded.

According to this constraint, only the low C layer, i.e., FinP, S.AspP and OnlyP, can be embedded, while the rest (Force and Attitude layers) can only appear in root clause. The same would be expected in LAC. Therefore, it is predicted that hū₁, yǔ₁, yǐ_{2b}, hū₂, yǔ₂, fū, and zāi only appear in root clauses, and never in embedded clauses. This prediction is indeed borne out: no appearances of these SFPs in embedded clause are observed. It is also predicted that the appearance of FinP, S.AspP and OnlyP heads, i.e., yě, yǐ₁, yǐ_{2a}, and ěr in embedded clauses will be observed.

Instances of embedded yě are observed, as in (29) below.

(29) 是可忍也，孰不可忍也？

[RheQP~ [CP [FinP shì kě rěn yě]] shù bù kě rěn yě]

This can tolerate YE what NEG can tolerate YE
 ‘If this can be tolerated, what cannot be tolerated?’ (Lau 2014 [2000]: 19)

(29) shows that the first *yě* appears in a conditional clause. This conforms to our prediction that *yě* as a FinP head can be embedded.

A few examples of embedded *yǐ*_{2a} are observed in the Analects, as in (30) and (31) below:

- (30) 吾必謂之學矣。
 [TP wú [MP bì [VP wèi zhī [S.AspP [TP xué] yǐ]]]
 1SQ must describe 3OBJ learn YI
 ‘I would grant that a man is, indeed, schooled [...]’ (Lau 2014 [2000]: 5)
- (31) 苟志於仁矣，無惡也。
 [FinP [TP [CP gǒu [S.AspP zhì yú rén yǐ]] wú è] yě]
 If set at benevolence YI no evil YE
 ‘If a man were to set his heart on benevolence, he would be free from evil.’ (Lau 2014 [2000]: 29)

In (30), the S.AspP head *yǐ*_{2a} is attached to the clause [TP xué] to form an embedded clause which is then selected by the light verb involved in the *wèi*-ditransitive construction. The ‘change of state’ meaning therefore contributes to the embedded clause, resulting in the meaning ‘has received instruction’/ ‘has already learned’. If it is attached to the main clause, the meaning of the sentence would be ‘I would already grant that a man learns x’, which is not the case here. In (31), *yǐ*_{2a} is embedded in the conditional clause to express a change of state: ‘once a man sets his heart on benevolence’. These two examples conform to our prediction that the S.AspP head *yǐ*_{2a} can be embedded in LAC.

On the other hand, embedded *yǐ*₁ and *ěr* cannot be found in the Analects. One example of semantically embedded *yǐ*₁ can be found in another source, the Book of Rites, as in (32) below.

- (32) 以死者為不可別已，故以其旗識之。
 [TP [CP yǐ [TP sǐzhě [VP wéi [S.AspP [TP bù kě bié] yǐ]]]]
 Because deceased COP NEG can differentiate YI
 [TP gù [VP [VP yǐ qí qí] [VP shí zhī]]]
 So use this flag identify 3OBJ
 ‘Because the appearance of the deceased can no longer be identified, this flag is used for identification.’⁵

In (32), the S.AspP head *yǐ*₁ is attached to the clause [TP bù kě bié yǐ] to form an embedded clause (which is under the scope of the copular verb *wéi* ‘be’) to express the meaning ‘can no longer be identified’. If the *yǐ*₁ is attached to the larger subordinate clause, the meaning of ‘already’ would scope over ‘because’, which is not the case here.

⁵ (32) and (33) are translated into English by the author based on the Modern Chinese translation in Wang (2009).

Embedded ẽr is not found in the Analects. This could be due to the paucity of ẽr in the Analects (only one sentence with the SFP ẽr is found, i.e., (6)). Again, one such example can be found in the Book of Rites, as in (33) below.

- (33) 何必小功耳？
 [IP Hé [MP bì [OnlyP [TP xiǎogōng] ẽr]]]
 Why must xiaogong ER
 ‘Why must it be only xiaogong (a type of mourning suit)?’

In (33), the OnlyP head ẽr is attached to the clause [TP xiǎogōng] to form an embedded clause under the scope of the modal verb bì ‘must’. The exclusive focus is confined to the embedded clause, meaning ‘only xiaogong’. If it is attached to the main clause, the meaning of the sentence would be ‘It is only that, why must it be xiaogong?’, which is not the case here.

Although the materials in the Book of Rites can be traced back to the Warring States period (475– 221 BC), the extant version of Book of Rites is believed to have been edited, selected, and compiled in Han dynasty (after 2nd c. BC) by the hands of different people such as Dai De (戴德) and Dai Sheng (戴圣) in the late Western Han, and Ma Rong (馬融) and Lu Zhi (盧植) in Eastern Han (Liu and Habberstad 2014). Book of Rites is possibly a mix of LAC from the 5th– 3rd c. BC and Chinese from the Han dynasty (202 BC–220 AD). Therefore, it is not reliable to treat linguistic data from the Book of Rites as a representative source for LAC.

Considering data only from the Analects, no example of embedded yǐ₁ and ẽr is found. This result does not align with our predictions from the Subjectivity Scale Constraint, which claims that these low C layer SFPs could be freely embedded. In sum, our predictions based on the Subjectivity Scale Constraint are only partially borne out: FinP head yě and S.AspP head yǐ_{2a} are found to appear in embedded clauses, while S.AspP head yǐ₁ and OnlyP head ẽr are not found to be embedded.

One explanation is that embedding of yǐ₁ and ẽr is possible but simply not covered by the limited linguistic data in the Analects. As aforementioned, ẽr only appeared once in the Analects, severely restricting the opportunity to observe its behaviour in embedded contexts. In general, embedded SFPs are possible but rare in LAC. Although instances of embedded FinP head yě and S.AspP head yǐ_{2a} are found, they are attested at low frequency. Other means to convey similar meanings and discourse functions might be preferred over embedding SFPs in LAC. For example, if LAC speakers want to express the meaning of ‘already’ in an embedded clause, adverbial clauses might be preferred. The reason behind this might be correlated with style or register.

Another explanation for the absence of attested embedded yǐ₁ and ẽr might be related to their subjectivity. Yǐ₁ and ẽr might not be purely S.AspP and OnlyP. They might contain an AttP part that prevents them from being embedded. Similar to Sybesma and Li’s (2007) approach to Cantonese SFPs, SFPs in LAC might also be decomposed into parts that head different projections. This might provide insights into why embedded yǐ₁ and ẽr are unattested in the Analects, and why they are restricted to root clause final position even when they are found embedded in the Book of Rites (as in (32) and (33)). However, such approach must be

examined carefully as decomposing SFPs into elements across the C layer might bring restrictions on their clustering. Future studies that aim to decompose SFPs in LAC c should be based on the reconstructed phonetic forms of the SFPs.

6. Conclusion

To conclude, this study starts out by assigning different functional projections to eight common SFPs in LAC. It is found that the eight characters represent eleven SFPs. A hierarchy of the eleven SFPs is constructed to account for their clustering and distribution. Moreover, yě, yǐ₁, yǐ_{2a}, and ěr are predicted to appear in embedded clauses according to the Subjectivity Scale Constraint. The results show that embedded yě and yǐ_{2a} are attested while embedded yǐ₁ and ěr are unattested in the Analects. Their absence might be attributed to the limited linguistic data in the Analects, and the rarity of embedded SFPs in LAC (albeit grammatical). Another possibility is that an AttP element within these SFPs restricts their embeddability. However, decomposing SFPs in LAC requires additional scrutiny in future studies.

7. References

- Cinque, Guglielmo. 1999. *Adverbs and functional heads: A cross-linguistic perspective*. Oxford University Press.
- Cinque, Guglielmo & Luigi Rizzi. 2009. The cartography of syntactic structures. In Vincenzo Moscati (ed.), *STiL – Studies in Linguistics*, Vol. 2. Siena: CISCL Press, 42–58.
- de Cat, Cécile. 2012. *Towards an interface definition of root phenomena*. In Lobke Aelbrecht, Liliane Haegeman and Rachel Nye (eds.), *Main Clause Phenomena: New Horizons*. Amsterdam: John Benjamins, 135–158.
- Cheng, Lisa Lai-Shen. 1991. *On the Typology of WH-Questions*. PhD dissertation, MIT.
- Djamouri, Redouane., and Waltraud Paul. 2019. Disharmony in harmony with diachronic stability: The case of Chinese. In *The determinants of diachronic stability* (pp. 101–130). John Benjamins Publishing Company.
- Lau, Din Cheuk. 2014 [2000]. *The Analects*. 1st paper edn. The Chinese University Press.
- Liu, Cheng-hui. 2007. The Functions of yi and its Semantic Extensions in Pre-Qin Chinese. *Language and Linguistics*, 8(3), 743–766.
- Liu, Yucai & Luke Habberstad. 2014. The Life of a Text: A Brief History of the Liji (Rites Records) and Its Transmission. *Journal of Chinese Literature and Culture*, 1(1-2), 289–308.
- Pan, Victor Junnan. 2021. *Architecture of the Periphery in Chinese*. Routledge.
- Rizzi, Luigi. 1997. The fine structure of the left periphery. In *Elements of grammar: Handbook in generative syntax* (pp. 281–337). Dordrecht: Springer Netherlands.

- Sybesma, Rint & Boya Li. 2007. The dissection and structural mapping of Cantonese sentence final particles. *Lingua*, 117(10), 1739–1783.
- Wang, Meng Ou. (王夢鷗). (2009). *Liji jin zhu jin yi* [Exegesis and translation of The Book of Rites]. Taiwan: Commercial Press.
- Wang, Aiqing. 2023. Exclamation in Late Archaic Chinese. *Acta Linguistica Asiatica*, 13(2), 59–85.
- Wei, Pei-chuan, P.M. Thompson, Chenghui Liu, Chu-Ren Huang & Chaofen Sun. 1997. Jiangou yi ge yi gongshi yu lishi yuyan yanjiu wei daoxiang de lishi yuliaoku [Historical Corpora for Synchronic and Diachronic Linguistics Studies]. *Int. J. Comput. Linguistics Chin. Lang. Process.*, 2.